

Placement, Routing and the Border Contract

What decides where a tenant's data lives, how a request reaches the cell that is allowed to hold it, and what may cross a national border. **This is a directional document:** a first-pass position recorded so the platform can move, not a settled decision, stated as at 3 September 2026. Nothing on this sheet is built — the lab runs one server in one region — and no V3 decision of record exists yet. Messaging mechanics belong to LB-V3-L3-05; the placement record, allow-list text and country rules belong to L4.

PLACEMENT POLICY · WHAT DECIDES WHERE A TENANT'S DATA LIVES

P **Tenant placement record**
names the permitted cells
held by the control plane
proposed · no V3 ADR

C **Country rule set**
what a country demands
residency, retention, transit
open · legal to confirm

D **Data class on each item**
the class decides the border
set where the data is made
proposed · no V3 ADR

A **One account, many cells**
region is not a tenancy axis
one identity in every cell
proposed · no V3 ADR

policy is resolved before any traffic is placed

ROUTING · HOW A REQUEST REACHES THE CELL THAT MAY HOLD ITS DATA

E **Country endpoint**
one entry name per country
HTTPS, ends in the country
proposed · no V3 ADR

R **Resolve before routing**
tenant and policy read first
where you are decides nothing
proposed · no V3 ADR

I **Cell ingress and web**
one door, one asset origin
shell and plugins are local
proposed · no V3 ADR

N **No silent re-routing**
an outage never moves data
a regulated request waits
proposed · no V3 ADR

the request lands in one cell, carrying its data class

DATA CLASSES · THE CLASS DECIDES WHAT MAY CROSS A BORDER

1 **Platform reference data**
shared, owned by nobody
may be copied to each cell
proposed · no V3 ADR

2 **Tenant business records**
loads, trips, invoices
stays inside its own cell
proposed · no V3 ADR

3 **Regulated country data**
named by a country rule
no border path at all
open · legal to confirm

4 **Operational telemetry**
metrics, traces, masked logs
crosses only by policy
open · legal to confirm

only a named path may take a class across a border

BORDER CONTRACT · THE ONLY THREE WAYS ANYTHING CROSSES

S **Signed policy and claims, in**
the control plane writes to every cell
policy, account claims and catalogue entries only
proposed · no V3 ADR · carries no business payload

G **Approved subjects, across**
a gateway between two shared cells only
an allow-list names every subject that may cross
open · which subjects, and whether at all

M **Named stream copy, across**
a mirror or source, never on by default
chosen because a copy resumes after a break
open · which streams, and in which direction

X **Sovereign border · nothing regulated crosses**
no gateway, no mirror, no backup copy, no remote support session
only signed policy and account claims enter, and they carry no payload
open · legal confirmation required before this is compliance evidence

! **Residency outranks convenience**
policy decides placement; the cell boundary enforces it
where a person signs in decides nothing about where data lives
directional position as at 2026-09-03 · no V3 ADR of record

Read it downward. Policy chooses the cell, routing delivers the request to it, the data class decides whether anything may leave, and the border contract lists the only three ways out. **Built today:** nothing on this sheet — the lab is one server in one region. **Not shown:** subjects, accounts and grants, which are LB-V3-L3-05 ; the record schema, allow-list text and country rules, which are L4 .

SCOPE	One question. Where may a tenant's data live, and what may cross a border? Availability and recovery are on sheet 2.
DIRECTION	Directional. A first pass as at 2026-09-03 so work can start. Not a decision, and not compliance evidence.
PLACEMENT	A placement record names the cells one tenant may use. It belongs to the tenant account, not to a person or a browser.
CLASS	Every item carries a data class . The class, not the service, decides what happens at a border.
ROUTING	The tenant is resolved first , then an ingress is chosen. Where a person happens to be decides nothing.
LOCAL	The page, the app shell and the plugin registry are served from the same cell as the data they show.
MIRROR	Durable history crosses only as a named copy . A copy is preferred over a live link because it resumes after a break.
SOVEREIGN	A sovereign cell has no regulated path out, in any direction. Backups and support sessions are included in that.
ACCOUNT	A tenant is one account in every permitted cell. The account carries the tenant; the deployment carries the region.
OPEN	Every amber tile needs a named owner to confirm it. Ten open items are listed in the L3-08 register .

LEGEND

Green boundary — the policy that decides placement
Solid tile — built in the lab; there is none on this sheet
Red panel — a path that is refused outright
Red barred line — a border nothing regulated crosses

Blue boundary — how a request reaches a cell
Dashed grey tile — proposed for V3, not built
Green line — a policy decision is applied
Arrow head — the direction of the relationship

Cyan boundary — the classes of data being placed
Dashed amber tile — open, a named owner must confirm
Blue line — a request travelling over HTTPS
Line label — names the relationship, never bare "uses"

Amber boundary — permitted cross-border paths
Amber panel — a rule that governs the whole sheet
Amber dashed line — data offered to a border rule

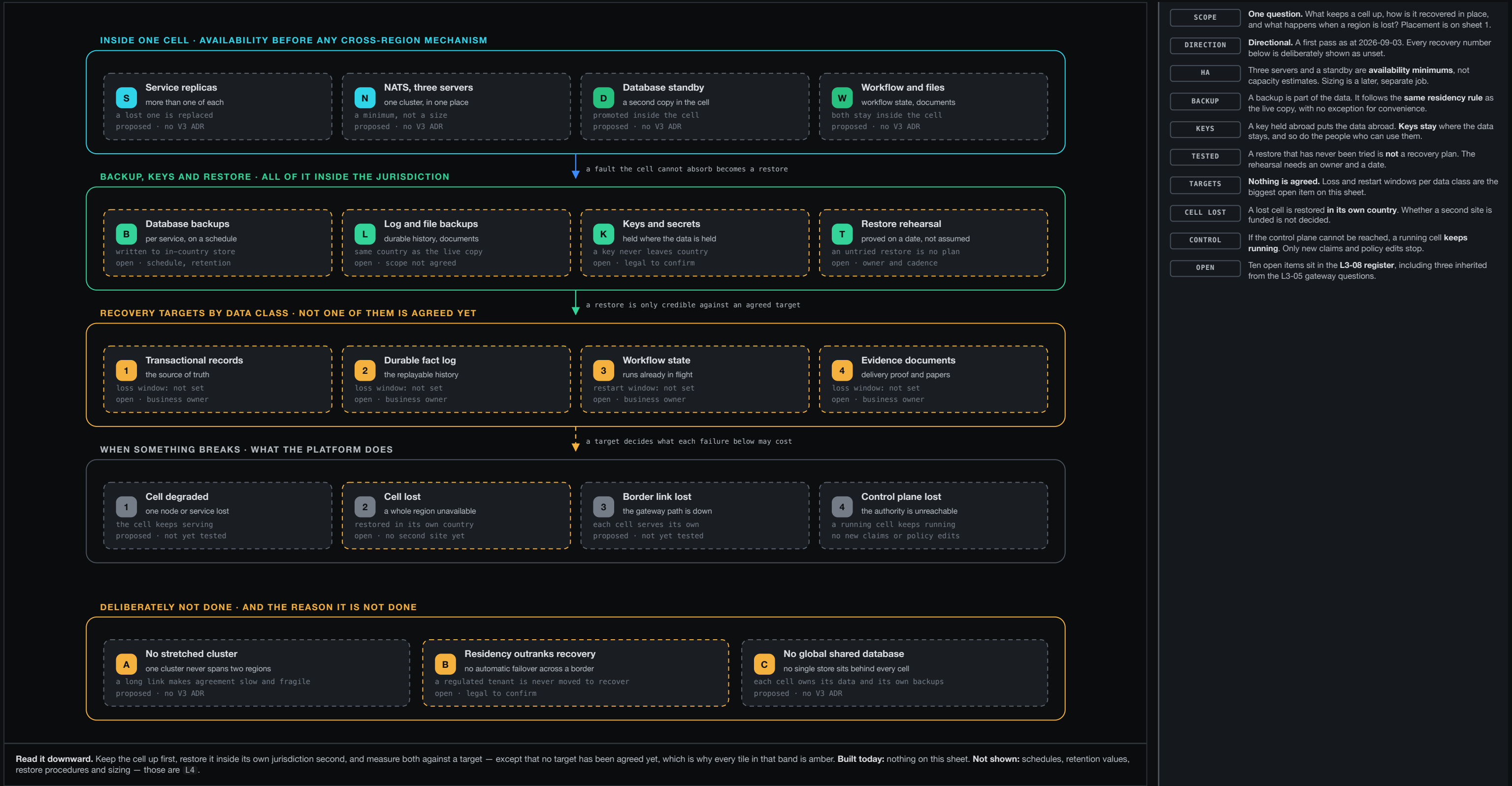
WORDS ON THIS SHEET

cell one region's complete runtime
region a place a cell is deployed in
tenant a marketplace-operating business
account the NATS tenant boundary
residency the rule saying where data may sit
data class a label deciding border behaviour
gateway a link joining two NATS clusters

mirror / source a copy of one stream elsewhere
sovereign kept inside one country by law
ingress the guarded door into a cell
HTTPS encrypted web traffic
ADR architecture decision record

Availability, Recovery and Failure by Jurisdiction

What keeps one cell running, how that cell is backed up and restored without leaving its jurisdiction, what each data class is allowed to lose, and what the platform does when a region, a border link or the control plane is lost. **This is a directional document:** a first-pass position stated as at 3 September 2026, not a settled decision. Nothing on this sheet is built, no recovery target has been agreed, and no V3 decision of record exists. Sizing, schedules and procedures belong to L4.



LEGEND

- | | | | | | | | |
|--|---|---|---|---|---|---|---|
| | Cyan boundary — availability inside one cell | | Green boundary — backup, keys and restore | | Amber boundary — unsettled, or deliberately refused | | Grey boundary — behaviour, not a component |
|  | Solid tile — built in the lab; there is none on this sheet |  | Dashed grey tile — proposed for V3, not built |  | Dashed amber tile — open, a named owner must confirm |  | Blue line — a fault escalating to the next answer |
|  | Green line — a restore measured against a target |  | Amber dashed line — a target governing a failure |  | Arrow head — the direction of the relationship | | Line label — names the relationship, never bare "uses" |

WORDS ON THIS SHEET

cell one region's complete runtime	rehearsal a restore tried on purpose, on a date
jurisdiction the country whose law applies	low window how much recent data may be lost
HA high availability, running through a fault	restart window how long a return may take
standby a second database copy, ready to take over	gateway a link joining two NATS clusters
promote making the standby the live copy	control plane the platform's own authority services
restore rebuilding from a backup	ADR architecture decision record